

DATA MANAGEMENT PLAN

1. Types of data generated

This project did not generate any raw field data and instead used existing breeding and climate datasets provided by third parties. Breeding monitoring data for the Southern Giant Petrel and Atlantic Yellow-nosed Albatross were provided by the Royal Society for the Protection of Birds and Tristan da Cunha Government. Climate data were provided by the South African Weather Service.

During this project, the data were cleaned and processed in R for analysis. Outputs from this included R scripts, candidate-window model comparisons, Bayesian model summaries and diagnostics, and figures, utilised in the dissertation.

2. Types of data preserved

The original breeding and climate datasets are not preserved or redistributed through the public GitHub repository because they are owned by third-party data providers and were supplied for this research on the conditions of confidentiality.

Materials and outputs generated through the analytical workflow that can be publicly shared are within the GitHub repository. These include R scripts, candidate-window comparison AIC outputs, Bayesian model summaries and diagnostics, and figures. These provide a record of the methodological process without redistributing the private raw data.

All raw or cleaned and processed datasets used for statistical analysis are not publicly available as this has not been authorised by the respective data owners.

3. Software and metadata implications

All data processing, statistical analysis and figure generation were conducted in R (version 4.6.0). Scripts are stored as .R files and annotated to provide context to the project's stages of cleaning, processing, statistical analysis, and figure generation. Outputs are preserved in commonly used formats, including .csv, .txt, and .png formats. README documentation within the GitHub repository describes the repository layout and purpose of the sub-directories. Variable naming is kept consistent throughout the workflow to support interpretation and reproducibility.

Limited details were provided regarding variables, monitoring protocols, and data sources for interpretation of methodology and results without access to restricted data. Relevant methodological information and citations to the original monitoring protocols and climate data sources are documented in the dissertation.

4. Length of data preservation

Research materials that are shareable will be retained in the GitHub repository following completion of the dissertation. This will remain publicly available long-term, but subject to the operation of GitHub and the maintenance of the repository by the account owner.

The raw and processed data will only be retained in a separate repository that is unavailable to the public, in accordance with the requirements of the data owners. This data is not available in the public repository used for assessment of the dissertation.

5. Value of data to others

The monitoring data provide insight into seabird reproduction, population dynamics and ecological drivers on Gough Island. These strengthen the case for continued monitoring to provide similar insight into future ecological and environmental analyses.

The publicly available analyses are valuable for demonstrating how the data were processed and analysed, supporting reproducibility of the study methodology. The annotated R scripts detail a transferable workflow for analysing climate associations with reproductive success using biologically defined candidate window analysis and Bayesian modelling. Model summaries, diagnostics, and subsequent figures provide detail and transparency regarding the outcomes of the analyses are reported in the dissertation.

6. Length of proprietary period

The original breeding and climate datasets are owned by third-party data providers and are not included in the public repository; access remains subject to the respective data owners. No proprietary period is applied to the materials generated during project analyses. R scripts model outputs, and figures are publicly available through the project GitHub repository to support the completion of the dissertation.

7. How data will be shared

Raw and processed data will not be shared through the public repository because the datasets are owned by third-party organisation. The repository instead clearly identifies the data sources and explains their restriction within the README. Research material that can be publicly distributed has been made available through the public dissertation GitHub repository:

<https://anonymous.4open.science/r/EEB-Dissertation-Public-C102/README.md>

This repository follows the FAIR principles wherever possible, by making the analytical materials findable through a documented repository, publicly accessible, interoperable through the use of common file formats, and reusable through annotated R code and documentation. Restrictions on the original underlying data have been documented to detail their origin but not their content.

8. Resources needed to preserve and share data

No specialist software is required to access or share the material within the GitHub repository. R and the R packages detailed in the README are required to reproduce the analytical workflow.